Minimal block III: the frozen coefficient theorem with noncritical coordinates

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Abstract

Freezing ξ, η on a minimal block of multiplicity $m = d + 1$ (they depend only on the noncritical coordinates v, whose exponents exceed the common block exponent p), the frozen state integral $Z_0(N) = \int_v v^{h'} e(v)\, Z_{d+1}^{a(v)}(Nv^{k'})\, dv$ satisfies $N^p Z_0(N)/\log^d N \to (\prod k_i^{-1}) \frac{1}{d!} \int_v v^{h'} (v^{k'})^{-p} A_{p-1}(a(v)) e(v)\, dv$. This is the multiplicity-m analogue of unit 70: dominated convergence over v with the uniform block envelope of unit 74 and the scale limit at $\gamma = v^{k'}$. Zero `sorry`/`axiom`.

1 The things formalised

```
noncomputable def frozenStateIntegral (  b N :  ) (k h :   →  ) (d :  ) {d' :  }
    (k' h' : Fin d' →  ) (a e : (Fin d' →  ) →  ) :   :=
   v : Fin d' →  , (  i, v i ^ h' i) * e v
    * iterChartGen   (a v) b k h (d + 1) (N *   i, v i ^ k' i)  (boxMeasure b d')
noncomputable def blockDivisorDensity (  p :  ) (k :   →  ) (d :  ) {d' :  }
    (k' h' : Fin d' →  ) (a e : (Fin d' →  ) →  ) (v : Fin d' →  ) :   :=
  (  i, v i ^ h' i) * e v * ((  i   Finset.range (d + 1), (1 :  ) / k i)
    * (1 / (d.factorial :  )) * (  i, v i ^ k' i) ^ (-p) * weightedMass   (a v) (p - 1))
theorem frozenStateIntegral_tendsto ... (hp :   i, i   d → ((h i :  ) + 1) / k i = p)
    (hq :   i, p < ((h' i :  ) + 1) / k' i) (hac : Continuous a) (hec : Continuous e) :
    Tendsto (fun N :   => N ^ p / Real.log N ^ d
        * frozenStateIntegral   b N k h d k' h' a e) atTop
      (  (  v, blockDivisorDensity   p k d k' h' a e v  (boxMeasure b d')))
```

2 Proof strategy

For $\log N \geq 1$ and v in the open box, the envelope gives $Z_{d+1}^{a(v)}(Nv^{k'}) \leq P\,E\,(1 + \log_+(Nv^{k'}B))^d (Nv^{k'})^{-p}$ and $\log_+(Nv^{k'}B) \leq \log N + \log_+(B'B)$, so the normalised integrand is dominated by $K\,v^{h'}(v^{k'})^{-p}$, integrable on the box because $q'_i > p$ (unit 71). The pointwise limit is the scale limit of unit 74 at $\gamma = v^{k'}$, multiplied by $v^{h'}e(v)$. Measurability in v comes from writing the block as a box integral over $(0, b]^{d+1}$ of a function jointly continuous in (v, u) and applying `StronglyMeasurable.integral_prod_right'`.

3 Roadblocks and resolutions

- `StronglyMeasurable` has no `congr` field (it unfolds to an existential); rewrite the function by `funext` first.

- Dot notation across a line break after a parenthesised term (`(foo …)\n .const_mul`) parses wrongly; write `Tendsto.const_mul c (foo …)`.

4 Outcome

Astra target 4 (frozen minimal-face coefficient) is done for a block of any multiplicity together with arbitrary noncritical coordinates. Remaining for the genuine leading theorem: the strip estimate for a block with one non-minimal coordinate, and the continuity freezing argument $Z - Z_0 = o(N^{-p} \log^d N)$.